

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Floating Point Representation

Lecture No.- 1

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Recap of Previous Lecture



Topic

Hardwired Control Unit

Topic

Microprogrammed Control Unit

Topic

Control Word Sequencing

Topics to be Covered



Topic

Floating Point Representation

Note :- If any signal can not be grouped with other signals in vertical micro-programmed control unit, then store them as horizontal way.

#Q. A micro-programmed control unit is required to support 128 instructions with each instruction need a sequence of 64 microoperations to execute completely. The control unit has microinstruction format with 3 fields: control signals, MUX select and next address field. The control unit has 134 signal to be stored in horizontal manner. The MUX has 8 inputs. The size of control memory to support such control unit is 150 kbytes?

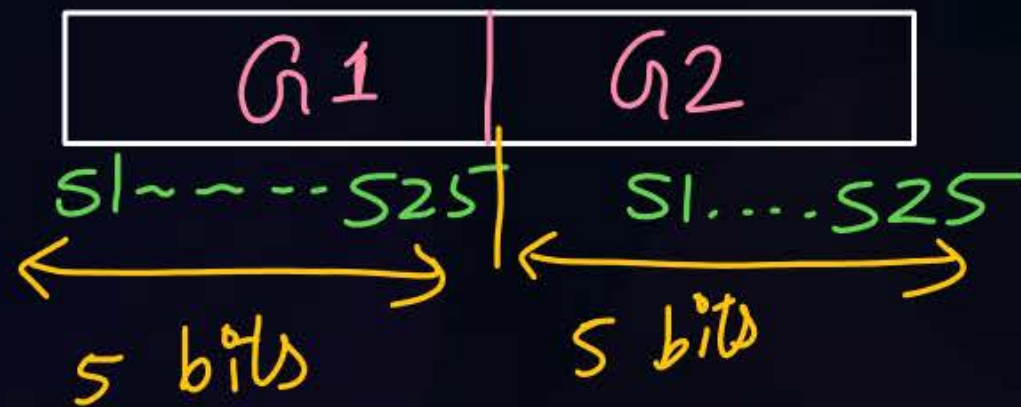
$$2^{13} * 150 \text{ bits}$$
$$150 \text{ k bytes}$$

Ans $\Rightarrow$ 10 bits

#Q. A micro-programmed control unit is required to generate a total of 25 control signals. Assume that during any microinstruction at most 2 control signals are active. Minimum number of bits required in the control word to generate the required control signals will be?

25 signals $\Rightarrow S_1, S_2, \dots, S_{25}$

make 2 groups for vertical microprogrammed C.U.





Topic : Floating-Point Numbers

→ for real numbers



floating point representation of numbers can represent a large range of number with limited number of bits.





Topic : Floating-Point Numbers

- The number is represented in format:



- Mantissa is signed normalized (implicit/explicit) fraction number
- Exponent is stored in biased form.

$$S = 1 \text{ bit} \begin{cases} 0 \Rightarrow +ve \\ 1 \Rightarrow -ve \end{cases}$$



Topic : Biased Exponent

S	E	M
---	---	---

← 4 bits →
↓ assume

largest possible range = -8 to $+7 \Rightarrow 0$ to 15
(2^5 complement)

E represented by	bias
4 bits	8
5 bits	16
6 bits	32
k bits	2^{k-1}

unsigned range

original exponent ^(e)	stored exponent (E) ← excess-8
-8	0
-7	1
-6	2
⋮	⋮
0	8
⋮	⋮
7	15

$$E = e + 8$$

$$\text{bias} = 8$$



Topic : Mantissa

101.1011

Explicit normalization

$$\Rightarrow 0.1011011 * 2^3$$

↓
always '1'

Implicit normalization

$$\Rightarrow 1.011011 * 2^2$$

↓ ↘
always 1 anything

$$e = 3$$

$$E = 3 + \text{bias}$$

$$m = \text{number after point} \\ = 1011011$$

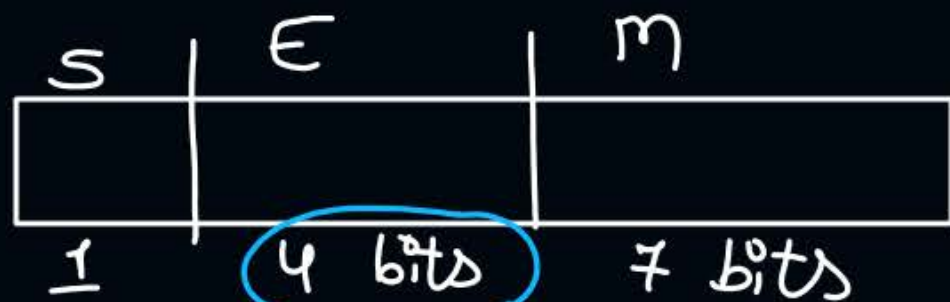
s	binary of 3 + bias	1011011
---	--------------------	---------

$$e = 2$$

$$E = 2 + \text{bias}$$

$$m = 011011$$

Example :-



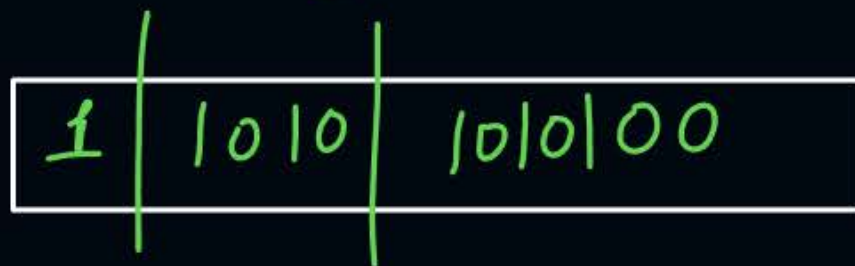
$$\text{bias} = 2^{4-1} = 2^3 = 8$$

explicit normalization used

$$\text{Number} \Rightarrow -(10.101)_2 \Rightarrow \text{Explicit normalization} \Rightarrow 0.10101 * 2^2$$

$$\text{negative} \Rightarrow S = 1$$

stored as



$$e = 2$$
$$E = 2 + 8 = (10)_{10} = (1010)_2$$

$$m = 10101$$

$$\Downarrow$$
$$1010100$$

To make it 7 bit mantissa add 2 zeros in end, because m is number after the point.



Topic : Value Formula

$$S=0 \quad (-1)^0 = 1 \Rightarrow +ve$$
$$S=1 \quad (-1)^1 = -1 \Rightarrow -ve$$

S	E	M
---	---	---

$$\text{Value}_{(\text{explicit})} = (-1)^S * 0.m * 2^{E-\text{bias}}$$

$$\text{Value}_{(\text{implicit})} = (-1)^S * 1.m * 2^{E-\text{bias}}$$

#Q. A certain well-known computer family represents the exponents of its floating-point numbers as "excess-64" integers; i.e., a typical exponent $e_6e_5e_4e_3e_2e_1e_0$ represents the number:

A $e = -64 + \sum_{i=0}^6 2^i e_i$

B $e = -64 + \sum_{i=0}^6 2e_i$

C $e = 64 - \sum_{i=0}^6 2^i e_i$

D $e = 64 - \sum_{i=0}^6 2e_i$

#Q. Consider a 16-bit register used to store floating point numbers. The mantissa is ^{explicit} normalized signed fraction number. Exponent is represented in excess-32 form. What is the 16-bit value for $+(11.5)_{10}$ in this register?

$$\text{bias} = 32 = 2^{k-1} \Rightarrow k = 6$$



$$S = 0$$

$$+ (1011.1)_2$$

explicit normalization

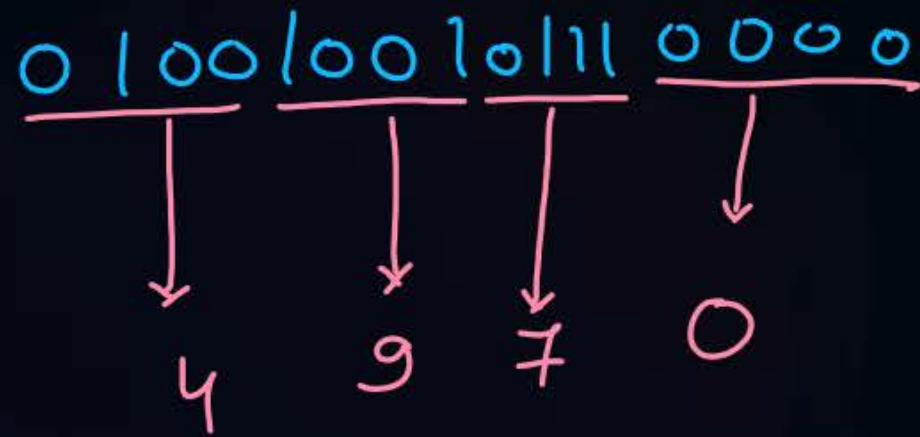
$$0.1011 \times 2^4$$

$$e = 4$$

$$E = 4 + 32 = 36 = (100100)_2$$

$$M = 101110000$$

#Q. What is the 4-digit hexadecimal value for $+(11.5)_{10}$ in above question's register?



$$(4970)_{16}$$

or

4970H

or

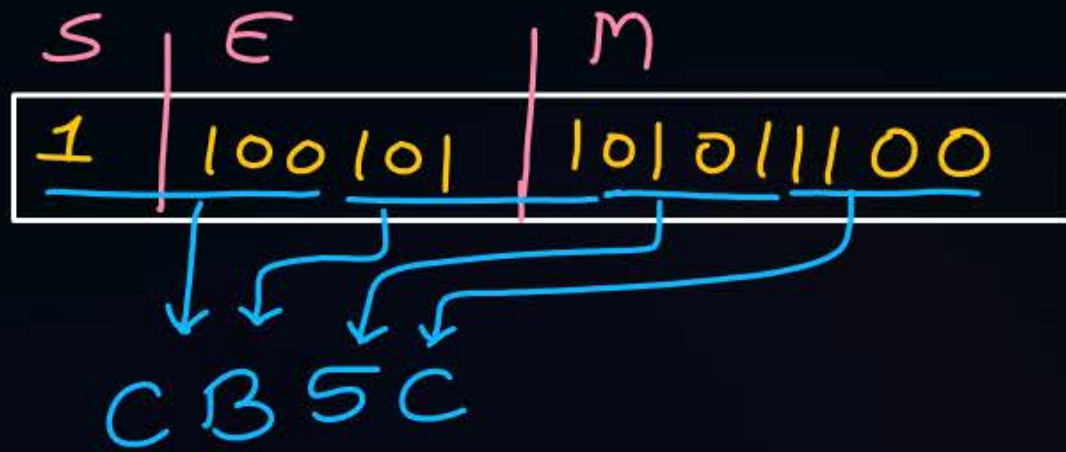
0x4970

[NAT]

Ans - CB5C



#Q. What is the 4-digit hexadecimal value for $-(21.75)_{10}$ in above question's register?



$$-(10101.11)_2$$

↓
Explicit normalization

$$0.1010111 * 2^5$$

$$e = 5$$

$$E = 5 + 32 = 37 = (100101)_2$$

$$m = 101011100$$

Underflow:-

In last questⁿ $E \Rightarrow 6$ bits

$$E_{\min} \Rightarrow 000000 = (0)_{10}$$

$$\text{original exponent } E_{\min} = 0 - 32 = (-32)_{10}$$

assume a number to store $\Rightarrow 0.\underbrace{00000\dots\dots 0}_{33 \text{ times zero}}11$

$\Downarrow$
normalize
 $\Downarrow$

$$0.11 * 2$$

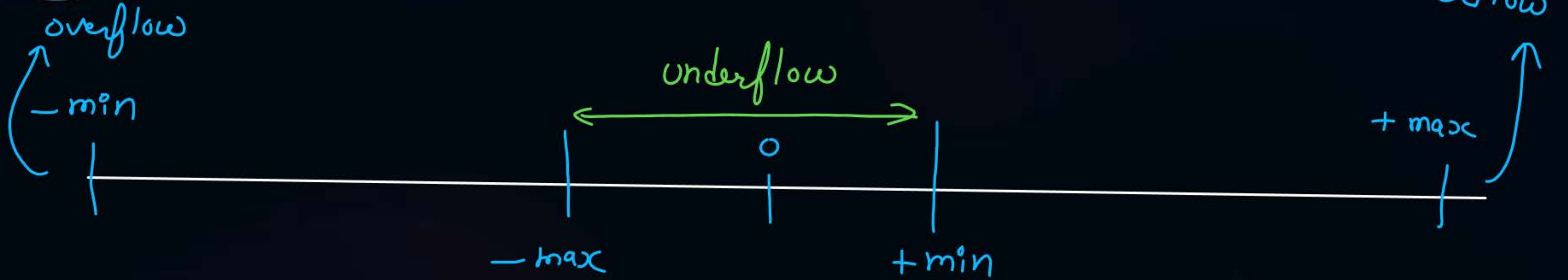
-33

$\Downarrow$ underflow

can not store it
because $e = -33$
which is smaller
than -32 .



Topic : Number Range





→ we can not store zero.

→ we can not store very small numbers around zero (underflow)

Implicit normalized $\Rightarrow \frac{1 \cdot m}{\text{always } 1}$

explicit $-11 \longrightarrow \Rightarrow 0.1 \dots$
 $\uparrow$ should 1



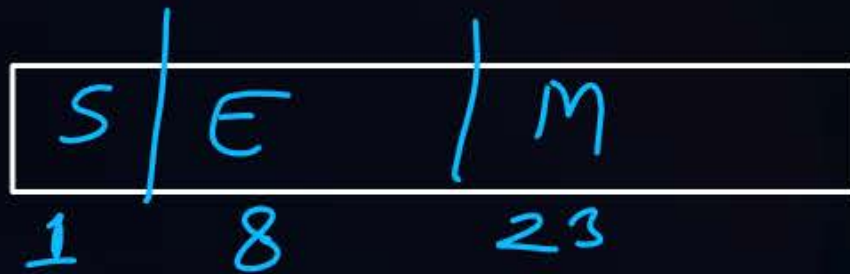
Topic : IEEE-754 Floating Point Representation

→ standard representation

IEEE-754
Representation

Single Precision

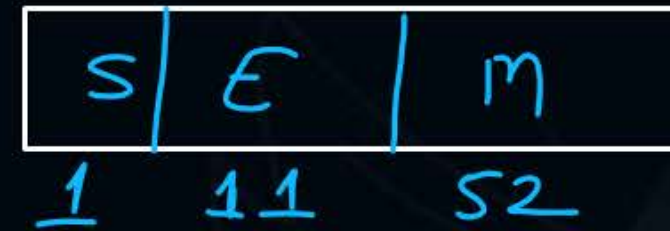
32 bits



bias = 127

Double Precision

64 - bits



bias = 1023

$E = 000\dots 0$
or
 $111\dots 1$ } $\Rightarrow$ special numbers

$E \neq 00\dots 0$
also
 $\neq 11\dots 1$ } $\Rightarrow$ normal number
(Implicit normalization)



2 mins Summary



Topic

Floating Point Representation





Happy Learning

THANK - YOU

